

Study of Power System State Estimation with Impact of PMU Considering Bad Data Measurement

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Abstract:

Monitoring and control of the power system have become very important functions, for achieving a high reliability in the system and improving the EMS functions. Recently, (PMU) has become one of the EMS applications for real time monitoring at the control center. In this paper, we used the Decoupled Current Measurement (DCM) method to include the (PMU) in State Estimation (ES) of Power System (PS). The proposed method has been tested on standard data IEEE 14-bus testing system as a case study under bad data conditions. MATLAB program software is used to realize the simulation process thus verifying the excellent performance of the proposed approach. The paper found, when the measurements bad data measurements, the system will lose its operating points, but the PMU can return the system operating point by reducing the percentages of average errors of voltage magnitude and voltage angle.

Key words: Phasor Measurement Units (PMUs); State Estimation (ES); Power System (PS); Decoupled Current Measurement (DCM); Energy Management System (EMS); IEEE: Institute of Electrical and Electronics Engineers.

مستخلص:

مراقبة والتحكم في نظام القدرة الكهربائية أصبحتا وظيفتين مهمتين جداً لتحقيق الثقة العالية في النظام وتحسين وظائف (EMS). مؤخراً، أصبحت وحدة قياس الطور (PMU) إحدى تطبيقات (EMS) لمراقبة الوقت الحقيقي في مركز التحكم. في هذه الورقة، تم استخدام تقنية قياسات التيارات (DCM) لتضمين (PMU) في تقدير حالة نظام القدرة، حيث تم تطبيق طريقة (DCM) على بيانات شبكة كهربائية قياسية مكونة من (IEEE 14-bus) كدراسة حالة تحت شروط البيانات السيئة. استخدم برنامج (MATLAB) لمحاكاة عملية البيانات السيئة في النظام للحصول على أفضل أداء لتقنية (DCM)، وجدت الورقة عند وجود بيانات سيئة مقاسه في القياسات، سيفقد النظام نقاط تشغيله ولكن (PMU) يمكن أن يرجع نقطة تشغيل النظام بتخفيض متوسط الخطأ المئوي لمقدار وزاوية الجهد الكهربائي.

الكلمات الدالة: وحدات قياس الطور؛ تقدير حالة؛ نظام القدرة الكهربائية؛ تقنية قياسات التيار، نظام إدارة الطاقة.

1. Introduction:

When using WLS and DCM estimation method, detection and identification of bad data are done only after the estimation process by processing the measurement residuals. The analysis is essentially based on the properties of these residuals, including their expected probability distribution. Bad data may appear several different ways depending upon the type, location and number of measurements that are in error. They can be broadly classified as: Single bad data, where only one of the measurements in the entire system will have a large error, and multiple bad data, where more than one measurement will be in error (Abur & Exposito, 2004, 30).

The PMU introduces new functionality measurement spectrum, with accurate time tagging and synchronization capability, and it is thus the next step in the technological evaluation of state estimation in electrical power systems. The factor that affecting the estimation quality is the measurement accuracy. Because the SCADA system doesn't have a high measurement accuracy

(Karlson, 2016). It is known that, SCADA system is used to measure the measurements data for the traditional state estimation (power flow, power injected on the buses and voltage magnitude) as mentioned before in (Abur & Exposito, 2004). Also in the mid-1980s, PMU was presented, it is very accurate than the SCADA device. It gives more accuracy for system state variables than the SCADA system. Hence, the PMUs become a very important device. Their utilizations significantly and safely increase not only for substation applications but also for the EMS applications in the control centers (Exposito, et al, 2011). The usage of PMU can directly measure both magnitude of the voltage and its angle at the bus where it is installed (Zivanovic & Cairns, 1996). Different methods have been proposed for installing PMU in state estimation. A new approach, Decoupled Current Measurement (DCM) method applied to study the influence of PMU in power system dynamic state estimation in (Dowi & Li, 2013), where the current measurement which is measured by PMU is decoupled into active and reactive measurement and added to the basic Weighted Least Squares (WLS) state estimation decoupled formula. Phasor Measurements in Dynamic State Estimation of Power Systems is proposed in (Jain & Shivakumar, 2009). An alternative method for study the utilization and benefits of PMUs for large scale power system state estimation is presented in (Yoon, 2005, 1). Role of State Estimation in Power System can be found in (Schweppe, 1985).

In this paper, a technique using DCM method to include PMUs in state estimation of power system is used. The impact of PMUs in state estimation of power system under bad data (Single and Multiple) conditions is explained. Also the effect of the number of PMUs, the effects of the PMUs locations and the weightage given to their measurements on the accuracy of estimation are investigated. A comparative study for the state estimation algorithm with and without PMU has been made. The proposed method is applied to the IEEE 14- bus test system as a case study (www.ee.washington.edu/research/pstca). The DCM is applied through two cases with and without PMU under bad data conditions test. In the first, a single bad data, when only one of the measurements in the entire system will have a large error. In the second, a multiple bad data, when more than measurements will be in error.

2. Previous Studies

Jain & Shivakumar, (2009) Presented a PMU based dynamic state estimation algorithm and studied the impact of PMUs on a dynamic state estimation technique and explored the effect of the number of PMUs, their location, and the weight given to their measurements on the accuracy of dynamic state estimation.

Dowi & Li, (2013) Discussed the Influences of Phasor Measurement Unit in Power System Dynamic State Estimation. This paper presents a new alternative approach for using the PMU in Dynamic State Estimation. Where, the rectangular coordinate of the current is decoupled and added to decoupled formulation of the Weighted Least Squares (WLS) state estimation.

3. Mathematical Model

This paper introduces a proposed method for adding PMUs in SE. Where, the rectangular coordinate of the current is decoupled and added to decoupled formulation of the Weighted Least Squares (WLS) state estimation. In this study, the impact of PMU in SE under bad data conditions has been explained for single and multi-bad data measurements in the entire system which have a large error has been studied. The proposed method is tested using standard data IEEE 14-bus system (www.ee.washington.edu/research/pstca).

3.1 Advantages & Drawbacks of the Weighted Least Squares Method:

Weighted Least Squares (WLS) has several advantages over other methods, including (Abur

& Exposito, 2004):

1. It is well suited to extracting information from small data sets.
2. It is the only method that can be used for data points of varying quality.

Disadvantages include:

1. It requires that you know exactly what the weights can have unpredictable results, especially when dealing with small samples. Therefore, the technique should only be used when your weight estimates are fairly precise. In practice, precision of weight estimates usually is not possible.
2. Sensitivity to outliers is a problem. A rogue outlier given an inappropriate weight could dramatically skew your results.

3.2 Conventional State Estimation:

State estimation techniques have been developed and used at the transmission level for over 30 years. The basic approach is called the Weighted Least Square (WLS) method. An excellent review of the state of the art can be found in (Wu, 1990). The mathematical model of WLS state estimation is based on the mathematical relations between the measurements and the state variables. Consider the set of nonlinear equation of the measurements:

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k \quad (1)$$

where $\mathbf{z}_k$ represent the measurement vector m -dimensional ($\mathbf{m} \times 1$), $\mathbf{x}_k$ is state variable dimensioned ($\mathbf{n} \times 1$), $\mathbf{h}(\mathbf{x}_k)$ is nonlinear function relating measurement to the state vector dimensioned ($\mathbf{m} \times 1$), $\mathbf{v}_k$ is measurements error with zero mean and standard deviation of $\mathbf{R}$, k is instant of time, m is the number of measurements and n is the number of state variables. In (Yoon, 2005).

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \quad (2)$$

Where $\mathbf{H} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \big|_{\mathbf{x} = \mathbf{x}^0}$ Jacobian matrix ($\mathbf{m} \times \mathbf{n}$)

The decoupled formulation of the WLS state estimation is given by:

$$[\mathbf{Z}_{A1}] = \begin{bmatrix} \mathbf{P}\mathbf{f}_{ij} \\ \mathbf{P}_i \end{bmatrix} \& [\mathbf{Z}_{R1}] = \begin{bmatrix} \mathbf{q}\mathbf{f}_{ij} \\ \mathbf{q}_i \\ \mathbf{V} \end{bmatrix} \quad (3)$$

$$\mathbf{Z}_{A1} = \mathbf{h}_{A1}(\boldsymbol{\theta}) + \mathbf{v}_{A1} \& \mathbf{Z}_{R1} = \mathbf{h}_{R1}(\mathbf{V}) + \mathbf{v}_{R1} \quad (4)$$

where the subscript "A" is an active power measurement and "R" is the reactive power measurement.

The Jacobian matrix can be written as:

$$\mathbf{H}_1 = \begin{bmatrix} \mathbf{H}_{AA1} & \mathbf{H}_{AR1} \\ \mathbf{H}_{RA1} & \mathbf{H}_{RR1} \end{bmatrix} \quad (5)$$

The decoupled gain matrix is given by:

$$\mathbf{G}_1 = \begin{bmatrix} \mathbf{G}_{AA1} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}_{RR1} \end{bmatrix} \quad (6)$$

where,

$$G_{AA1} = H_{AA1}^T R_{A1}^{-1} H_{AA1} \text{ and } G_{RR1} = H_{RR1}^T R_{R1}^{-1} H_{RR1}$$

$$\Delta \theta = G_{AA1}^{-1} H_{AA1}^T R_{A1}^{-1} \left| Z_{A1} - h_{A1}(\tilde{\theta}) \right| \quad (7)$$

$$\Delta V = G_{RR1}^{-1} H_{RR1}^T R_{R1}^{-1} \left| Z_{R1} - h_{R1}(\tilde{V}) \right| \quad (8)$$

3.3 Decoupled Current Measurement (DCM) Method

In this method, the decoupled formulation of Weighted Least Square (WLS) is used in (Hongbo, et al, 2012); (Dowi, 2015). The current measurement which is measured by PMU is decoupled into active and reactive measurement and added to the WLS state estimation decoupled formula. Hence, the new formulation of the measurements set is written as:

$$\left[Z_{A2} \right] = \begin{bmatrix} Z_{A1} \\ \theta_i \\ I_{(ij)r} \end{bmatrix} \quad \& \quad \left[Z_{R2} \right] = \begin{bmatrix} Z_{R1} \\ V_i \\ I_{(ij)i} \end{bmatrix} \quad (9)$$

The subscripts "r" and "i" represent the real and imaginary parts of the PMUs respectively. The $I_{(ij)r}$ is added to active measurements, and $I_{(ij)i}$ added to the reactive measurements. Consider the apparent power where, $S_{ij} = P_{ij} + jQ_{ij}$, that mean, $V_i I_{ij} = V_i I_{ij} \cos \theta_{ij} + j V_i I_{ij} \sin \theta_{ij}$, assume that the voltage magnitude approximately equal to 1.0p.u then, $I_{ij} = I_{ij} \cos \theta_{ij} + j I_{ij} \sin \theta_{ij} = I_r + j I_i$, where $I_r = I_{ij} \cos \theta_{ij}$ & $I_i = I_{ij} \sin \theta_{ij}$.

The line current I_{ij} in line "ij" is calculated according to equations (17) & (18) as in (Jain & Shivakumar, 2009). The admittance of the series line connecting buses "i" and "j" is $y_{ij} = g_{ij} + j b_{ij}$, and the shunt branch admittance at bus "i" is $y_{i0} = j b_{i0}$. The new nonlinear equation of the measurements is obtained by the following.

$$Z_{A2} = h_{A2}(\theta) + v_{A2} \text{ \& } Z_{R2} = h_{R2}(V) + v_{R2} \quad (10)$$

The new Jacobian matrix can be written as:

$$H_2 = \begin{bmatrix} H_{AA2} & H_{AR2} \\ H_{RA2} & H_{RR2} \end{bmatrix} \quad (11)$$

The decoupled gain matrix is given by:

$$G_2 = \begin{bmatrix} G_{AA2} & 0 \\ 0 & G_{RR2} \end{bmatrix} \quad (12)$$

Where,

$$G_{AA2} = H_{AA2}^T R_{A2}^{-1} H_{AA2} \text{ \& } G_{RR2} = H_{RR2}^T R_{R2}^{-1} H_{RR2}$$

$$\Delta \theta = G_{AA2}^{-1} H_{AA2}^T R_{A2}^{-1} \left| Z_{A2} - h_{A2}(\tilde{\theta}) \right| \quad (13)$$

$$\Delta V = G_{RR2}^{-1} H_{RR2}^T R_{R2}^{-1} \left| Z_{R2} - h_{R2}(\tilde{V}) \right| \quad (14)$$

The estimated error's covariance matrix is given by:

$$\Sigma_{A2} = \left| H_{AA2}^T W_{A2} H_{AA2} \right|^{-1} \quad (15)$$

$$\Sigma_{R2} = \left[H_{RR2}^T W_{R2} H_{RR2} \right]^{-1} \quad (16)$$

where $\mathbf{W}$ is the weighted matrix of both SCADA and PMU, it is block diagonal matrix.

$$\mathbf{W}_{A2} = \begin{bmatrix} \mathbf{W}_{A2_{SCADA}} & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_{A2_{PMU}} \end{bmatrix} = \mathbf{R}_{A2}^{-1} \quad (17)$$

$$\mathbf{I}_{(ij)r} = g_{ij} (V_i \cos \theta_i - V_j \cos \theta_j) - b_{ij} (V_i \sin \theta_i - V_j \sin \theta_j) - b_{i0} V_i \sin \theta_i \quad (18)$$

$$\mathbf{I}_{(ij)i} = b_{ij} (V_i \cos \theta_i - V_j \cos \theta_j) + g_{ij} (V_i \sin \theta_i - V_j \sin \theta_j) + b_{i0} V_i \cos \theta_i \quad (19)$$

The Jacobin matrix of the phasor measurement is written as.

For active measurements:

$$\mathbf{H}_A = \begin{bmatrix} \frac{\partial \theta_i}{\partial \theta_i} & \frac{\partial \theta_j}{\partial \theta_j} \\ \frac{\partial \mathbf{I}_{(ij)r}}{\partial \theta_i} & \frac{\partial \mathbf{I}_{(ij)r}}{\partial \theta_j} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{C}_1 & \mathbf{C}_2 \end{bmatrix} \quad (20)$$

For reactive measurements:

$$\mathbf{H}_R = \begin{bmatrix} \frac{\partial V_i}{\partial V_i} & \frac{\partial V_j}{\partial V_j} \\ \frac{\partial \mathbf{I}_{(ij)i}}{\partial V_i} & \frac{\partial \mathbf{I}_{(ij)i}}{\partial V_j} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{C}_3 & \mathbf{C}_4 \end{bmatrix} \quad (21) \text{ Where:}$$

$$\mathbf{C}_1 = \frac{\partial \mathbf{I}_{(ij)r}}{\partial \theta_i}, \mathbf{C}_2 = \frac{\partial \mathbf{I}_{(ij)r}}{\partial \theta_j}, \mathbf{C}_3 = \frac{\partial \mathbf{I}_{(ij)i}}{\partial V_i}, \mathbf{C}_4 = \frac{\partial \mathbf{I}_{(ij)i}}{\partial V_j}$$

4. The Simulation analysis:

4.1 Description of simulation

In this paper, to validate the effectiveness of the proposed method, IEEE 14-bus test system has been used and tested under bad data conditions. The measurement vector consists of 75 observations. The measurements error covariance $\mathbf{R}$ for both SCADA and PMU are fixed during simulation experiments. The SE of the system is investigated over a period of 20 time sample intervals. During each interval, the load per bus is increased at a constant value of %5 for all buses with a constant power factor. Consequently, the reactive power will flow the real power.

4.2 Performance Indices

The performance index of the algorithm is obtained by comparing the estimated values with the available actual values. The average performance indices for voltage magnitudes & voltage angles is given as:

$$\epsilon_{v_Estimated} = \frac{1}{n} \sum_{i=1}^n \left| \frac{(V'_{i_actual} - V'_{i_estimated})}{V'_{i_actual}} \right| \times 100\% \quad (22)$$

$$\epsilon_{\theta_Estimated} = \frac{1}{n} \sum_{i=1}^n \left| \frac{(\theta'_{i_actual} - \theta'_{i_estimated})}{\theta'_{i_actual}} \right| \times 100\% \quad (23)$$

Where:

$\epsilon_{v_Estimated}$ is the percentage error estimated of voltage magnitude.

$\epsilon_{\theta_Estimated}$ is the percentage error estimated of voltage angle.

V'_{i_actual} is the actual voltage magnitude.

θ'_{i_actual} is the actual voltage angle.

$V'_{i_estimated}$ is the estimated voltage magnitude.

$\theta'_{i_estimated}$ is the estimated voltage angle.

4.3 Results and Analysis:

In this paper, the proposed approach is applied through two cases for single and multi-bad data measurements. The performance indices of this subsection shown in the tables and figures below. Results are represented and discussed below.

Table 1: Voltage magnitude V4 at bad data conditions

Time Sample	Actual	Estimated	
		Without PMU	With PMU
1	1.0174	1.0157	1.0177
2	1.0173	1.0130	1.0176
3	1.0173	1.0245	1.0177
4	1.0173	1.0185	1.0171
5	1.0173	1.0192	1.0174
6	1.0173	1.0232	1.0165
7	1.0173	1.0216	1.0180
8	1.0173	1.0186	1.0172
9	1.0173	1.0129	1.0175
10	1.0173	1.0201	1.0171
11	1.0174	1.0161	1.0170
12	1.0173	1.0171	1.0177
13	1.0173	1.0153	1.0179

14	1.0173	1.0171	1.0175
15	1.0173	1.0140	1.0172
16	1.0173	1.0113	1.0179
17	1.0173	1.0122	1.0172
18	1.0173	1.0168	1.0171
19	1.0173	1.0106	1.0179
20	1.0173	1.0209	1.0172

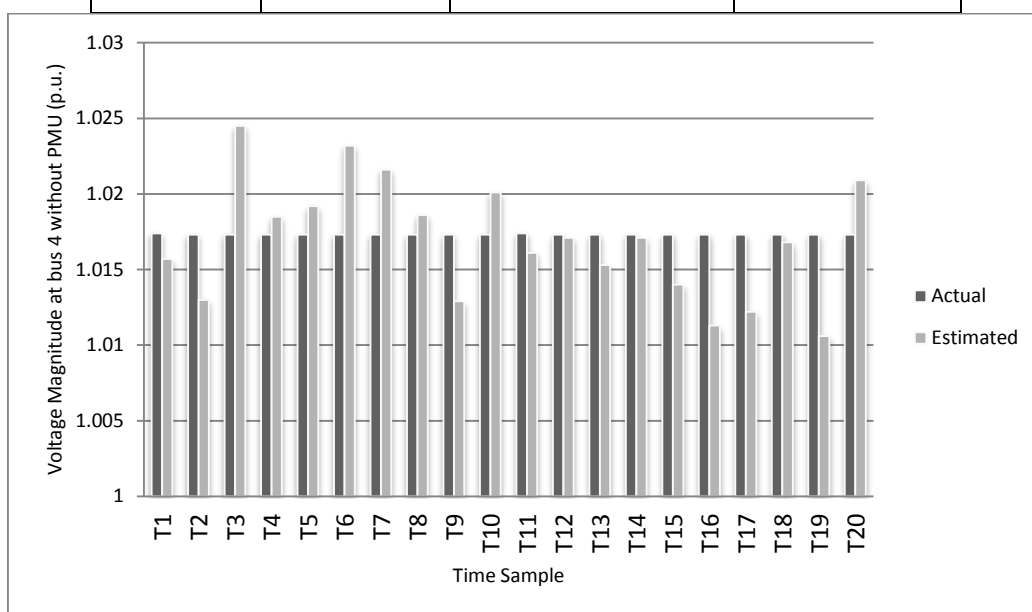


Figure 1: Voltage magnitude V_4 under bad data without PMU

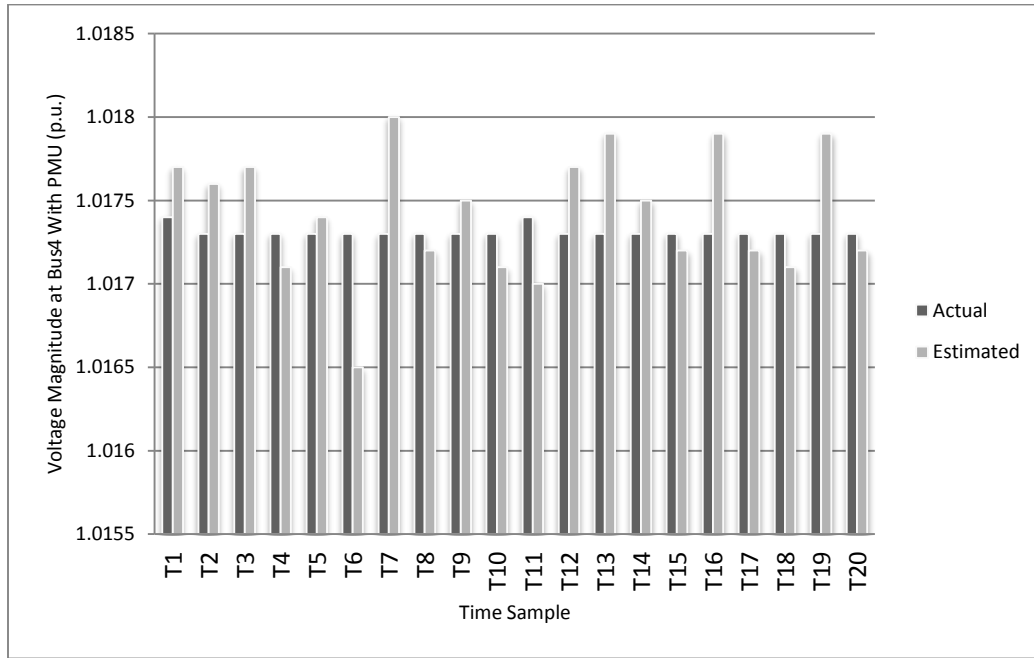


Figure 2: Voltage magnitude V_4 under bad data with PMU

Table 1, Figure 1 and Figure 2 represent the behaviors of the actual and estimated values for voltage magnitude under single bad data condition without and with PMU. The magnitude of the oscillations with PMU is better than without PMU.

4.3.1 Single Bad Data Case

In this case, the proposed method is applied to the scenario where single bad data are included in power system measurement, where, the reactive power injection $Q(4)$ was suspected in error of %50 increase at the 11th time sample. Suppose that no work is taken to eliminate these errors. Table 2, Figure 3 and Figure 4 show the various locations of individual PMUs and the corresponding weight vectors, which give the least error at that location under single bad data condition.

Table 2: Average estimated error (%) under single bad data conditions

Location of PMU	Weight	Estimated error (%)	
		Voltage	Angle
No PMU	-	0.5657	0.1394
2	100	0.1063	0.0256
3	1000	0.1047	0.0354
4	200	0.0572	0.0242
5	100	0.0974	0.0291
6	100	0.0521	0.0245

7	1000	0.0312	0.0155
8	1000	0.0881	0.0210
9	100	0.0517	0.0193
10	1000	0.1048	0.0185
11	1000	0.0661	0.0133
12	1000	0.0916	0.0274
13	1000	0.0759	0.0190
14	1000	0.0769	0.0142

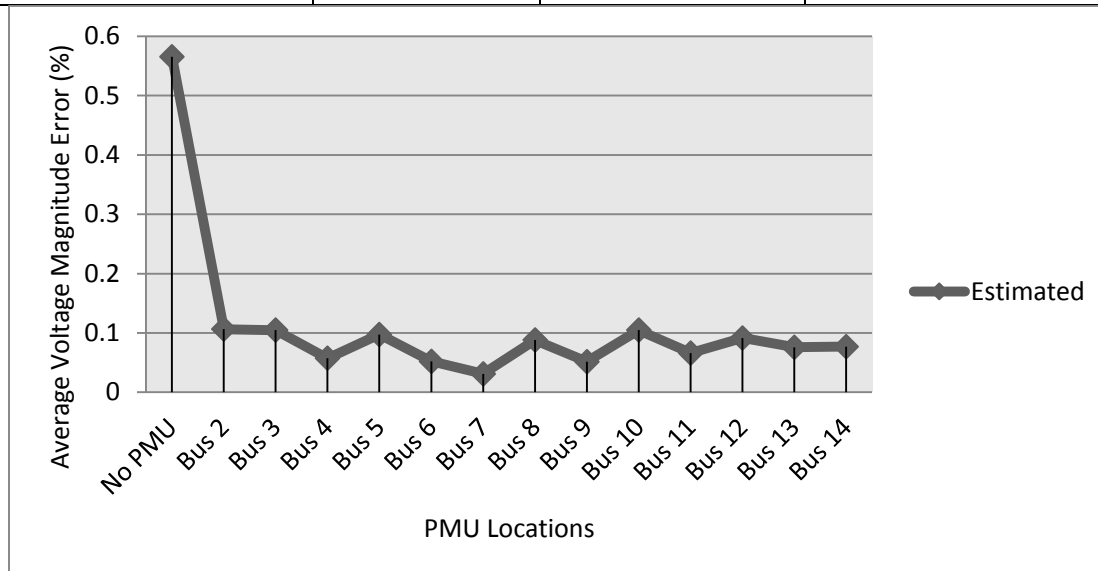


Figure 3: Average errors of voltage magnitude under single bad data conditions

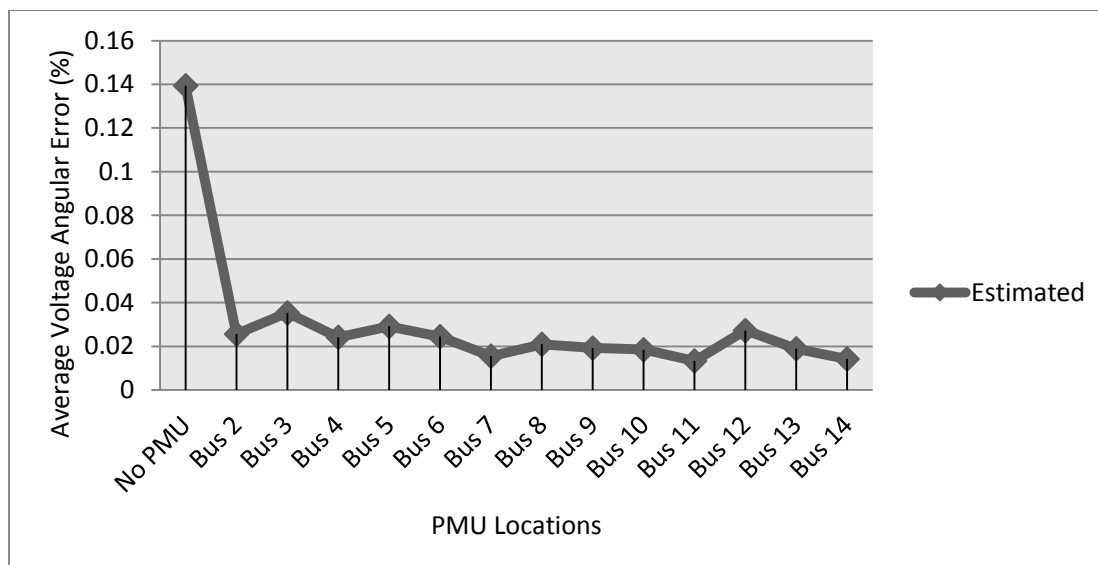


Figure 4: Average errors of voltage angle under single bad data conditions

Table 2 shows the performance indices of the proposed method (DCM method), where, about (%81 - %98) reduction of the voltage magnitude error, and about (%75 – %90) reducing of estimated voltage angular error is introduced.

From Table 2 we can clearly see the degree of the high reduction in the error. There is significant improvement in the estimated state. This advantage has a positive effect on the accuracy of security analysis. Furthermore, a high reduction of the error from estimated state shows the accuracy of the PMU fixed single bad data measurements. From all these positive impacts, we can conclude that the PMU has high ability to absorb the single bad data measurements. Also in this case a weightage of 1000 is ideal for achieving good estimates error, when the PMU placed at bus 7 (The best average voltage magnitude estimate error %0.0312) and bus 11 (The best average voltage angular estimate error %0.0133). Also the impact of the PMU is very clear for reducing the percentage of error of the estimated state at all buses under single bad data measurement.

Figure 3, and Figure 4, represent the variation of the average estimated error when the PMU is placed at each bus under single bad data condition and show the minimum and maximum average value of the estimated error found at different buses. From Figure 3, the best estimated error for the voltage magnitude due this single bad data measurement is obtained when the PMU is placed at bus 7, and the worst one is obtained when the PMU is placed at bus 2. From Figure 4, the best estimated error for the voltage angular is obtained when the PMU is placed at bus 11, and the worst one is obtained when the PMU is placed at bus 3. That means, the estimated error is affected by the location of the PMU under single bad data conditions and the location of the PMU is very important for estimating the error. From all these positive impacts, we can conclude that the DCM approach is very useful to deal with single bad data measurements and the PMU has high ability to absorb the single bad data.

4.3.2 Multiple Bad Data Case:

In this test, the simulation was carried out under multiple bad data conditions, for this scenario, the active power flow Pf (3,4) and reactive power injection Q (3) was suspected in error of %30 decrease and %40 increase respectively at 16 time sample. Suppose that no work is taken to eliminate these errors. Table 3, Figure 5 and Figure 6 show the various locations of individual PMUs and the corresponding weight vectors, which give the least error at that location under multiple bad data condition.

Table 3: Average estimated error (%) under multiple bad data conditions

Location of PMU	Weight	Estimated error (%)	
		Voltage	Angle
No PMU	-	0.6087	0.4433
2	100	0.0525	0.0417
3	1000	0.0292	0.0261
4	200	0.0728	0.0278
5	100	0.2819	0.1157
6	100	0.1854	0.1229
7	1000	0.1172	0.0508
8	1000	0.2141	0.0589

9	100	0.1304	0.0786
10	1000	0.2276	0.0539
11	1000	0.1868	0.0568
12	1000	0.3124	0.1471
13	1000	0.2579	0.1123
14	1000	0.1934	0.0476

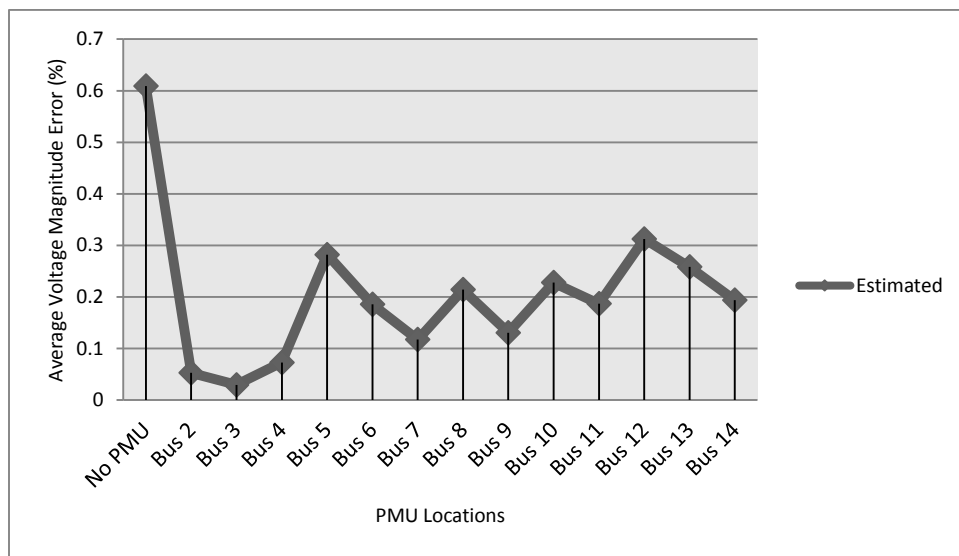


Figure 5: Average errors of voltage magnitude under multiple bad data conditions

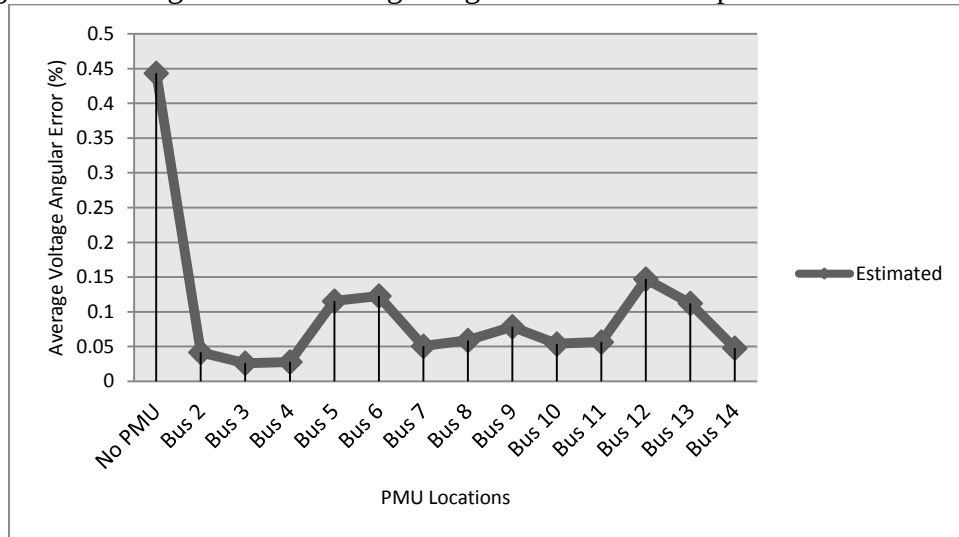


Figure 6: Average errors of voltage angle under multiple bad data conditions

From the Table 3, we can clearly see, different weight values to the PMU measurements give different accuracies to the estimated values. For the test case considered here, it was found that

the weightage 1000 provided estimates of good accuracy, when the PMU placed at bus 3 gives the best average voltage magnitude estimate error %0.0292 and the best average voltage angular estimate error %0.0261. Also the impact of the PMU is very clear for reducing the percentage of error of the estimated state at all buses under multiple bad data measurement. This reduction represents (%47 - %95) for voltage magnitude and (%69 – %94) for voltage angular.

Figure 5, and Figure 6, show the variation of the average estimated error when the PMU is placed at each bus under multiple bad data conditions. Also from Figure 5, the best estimated error for the voltage magnitude is obtained when the PMU is placed at bus 3, and the worst one is obtained when the PMU is placed at bus 12. From Figure 6, the best estimated error for the voltage angular is obtained when the PMU is placed at bus 3, and the worst one is obtained when the PMU is placed at bus 12. That means, the estimated error is affected by the location of the PMU under multiple bad data conditions and the location of the PMU is very important for estimating the error. Also, from Figure 5 and Figure 6, it is observed that most of the estimated values with PMU are much lower than the values without PMU. This feature shows the accuracy of the proposed approach, and it has a high effect for the security analysis at the control center. Furthermore, will reduce the security risk. Hence, it increases the accuracy and reliability to the system.

5. Conclusion:

The implementation of PMU in SE technique has been presented. The technique used the DCM method to include PMU in SE. The results ensure the importance of using PMUs in SE. From the results in single and multiple bad data conditions, it can be observed that the estimated values with PMU are much lower than the estimated values without PMU, this feature has a positive impact of PMU in SE. Furthermore, it shows the importance of the location of the PMU to achieve the best estimation and the weightage given to the PMU measurements also affects the accuracy of the estimates, as well as the capability of the PMUs to absorb the single and multiple bad data measurements. Also we can conclude that the PMUs can improve the bad data detection and identification. For further research, an enhancement on this study is proposed by performing the required simulation for large and practical systems with real measurements data.

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